

LECTURE ACTIVITY 8: FACTORS THAT AFFECT p -VALUE – Solutions

STAT 1110: Applied Statistical Concepts and Methods – Fall 2026 with Dr. Ethan P. Marzban

This activity is heavily adapted from one created by the STAT 1110 Course Coordination Team, which in turn is modified from “Introduction to Statistical Investigations,” by Tintle et. al.

Learning Objectives:

By the end of this activity, you should be able to

- 1) Identify two factors and how they affect strength of evidence against the null hypothesis.
- 2) Write the alternative hypothesis for a two-sided test.
- 3) Calculate a two-sided p -value.

Key Terms:

- Two-sided alternative
-

Recap

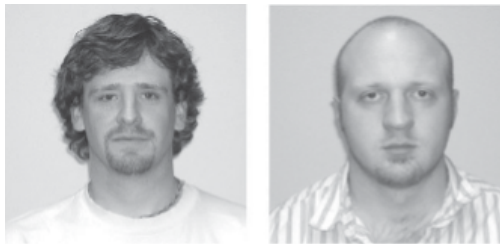


Figure 1: Melissa A. Lea/Robin D. Thomas/Nathan A. Lamkin/Aaron Bell ©Psychonomic Bulletin & Review 2007, 14 (5), 901–907.

Recall the setup of our last lecture activity, where we wanted to determine whether or not students have a tendency to (are more likely to) associate certain facial features with a name. I presented the two faces shown in Figure (1) to you (my three STAT 1110 lecture sections), and asked which name you would assign to the photo on the left. Of the 68 respondents, 52 assigned the name “Tim” to the photo on the left.

The parameter of interest is π_{TimLeft} , the population proportion of all STAT 1110 students that would assign the name “Tim” to the photo on the left.

As a quick aside: we technically could have also used the parameter “the long-run proportion of all STAT 1110 students that would assign the name “Tim” to the photo on the left.” However, to be consistent with our work in the previous lecture activity, let’s continue using the population proportion as our parameter of interest.

The null and alternative hypotheses are given, respectively, by

$$\begin{cases} H_0 : \pi_{\text{TimLeft}} = 0.5 \\ H_A : \pi_{\text{TimLeft}} > 0.5 \end{cases}$$

Additionally, shown below in Figure (2) is a screenshot from the [One Proportion Applet](#) carrying out a **simulation-based** test of the above-stated hypotheses

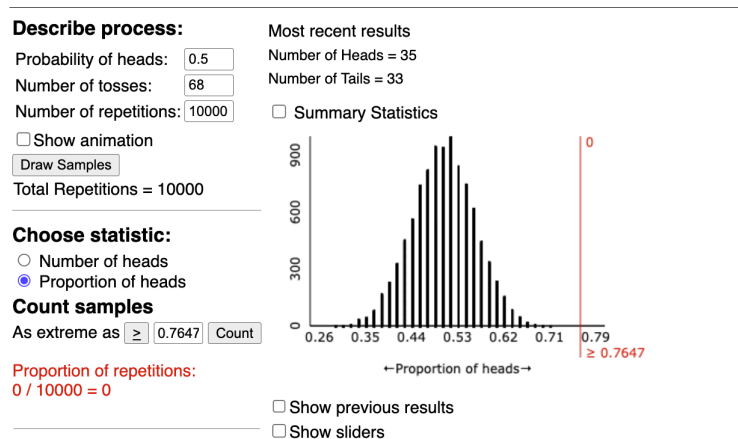


Figure 2: 10,000 repetitions of the chance model, as obtained from the One Proportion Applet

The approximate p -value obtained from the simulation-based approach is 0.000. Furthermore, since the **validity conditions** are satisfied, we can also use a **theory-based approach** to obtain an approximate p -value; the resulting theory-based approximate p -value, as obtained from the **Theory-Based Inference applet**:

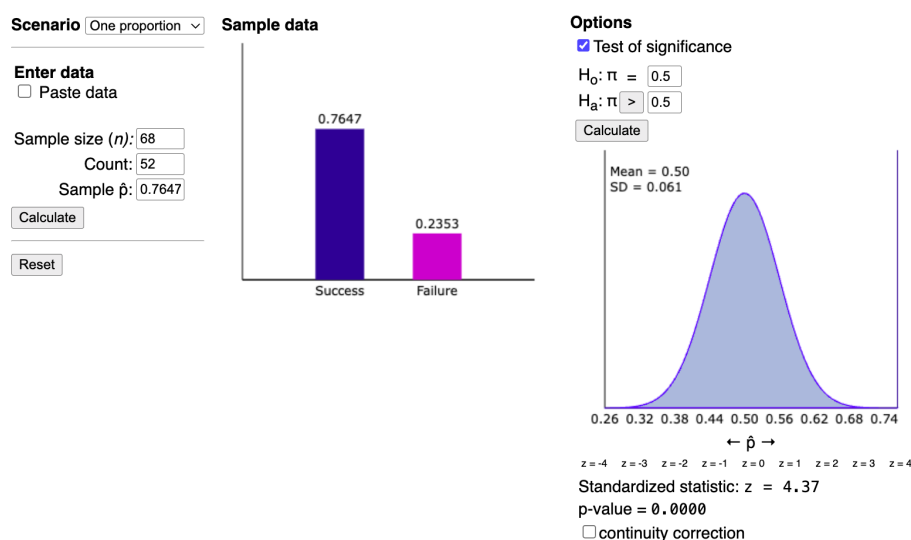


Figure 3: The theory-based approximate p -value, obtained from the Theory-Based Inference Applet.

Furthermore, our standardized statistic value is around $z = 4.37$ under both the simulation-based and theory-based approaches.

A fully-specified conclusion to this study would read:

Based on the approximate p -value of 0.000, we have very strong evidence against the null hypothesis leading us to reject it. We have sufficient evidence that the population proportion of students who would assign the name "Tim" to the picture on the left is more than 0.50. In other words, we have sufficient evidence that students are more likely to assign the name Tim to the face on the left. The result of this study apply to STAT students similar to the ones in the study.

Part I: Factors that Affect the Strength of Evidence Against H_0

In this part of the activity, we are going to explore two factors that influence the strength of evidence in a test of significance: (A) the difference between the observed sample statistic and the value of the parameter used in the null hypothesis and (B) the sample size

(A) Difference Between Statistic and Null Hypothesis Parameter Value

A factor that influences the strength of evidence against the null is how far apart the observed sample statistic and the value of the parameter specified under the null hypothesis are. For this study the null value was 0.50 and the observed sample statistic was about 0.765 (or 76.5% of the students chose Tim on the left). Suppose that instead of 52 out of the 68 students assigning “Tim” to the left, we had only 41 students assign “Tim” to the left (for a sample proportion of around 60.3%).



Question 1

- (a) How do you think changing the sample proportion to be a smaller number (0.603 instead of 0.745) will affect the approximate p -value? (This question is asking for your opinion.)
- ☐ The approximate p -value will be smaller
 - ☐ The approximate p -value will stay the same
 - ☐ The approximate p -value will be larger

Solution: There isn't really a right or wrong answer to this part - it's just asking for your opinion!

- (b) How do you think changing the sample proportion to be a smaller value (0.603 instead of 0.745) will affect the strength of evidence against the null hypothesis? (This question is asking for your opinion.)
- ☐ The evidence will be stronger
 - ☐ The evidence will stay the same
 - ☐ The evidence will be weaker

Solution: Again, there isn't really a right or wrong answer to this part - it's just asking for your opinion! But, I do encourage you to make sure is consistent with your answer to the previous part - remember that the smaller the p -value, the stronger the evidence against the null (and, on the flipside, the larger the p -value, the weaker the evidence against the null)



Question 2

Use the Theory-Based Inference Applet to calculate the new p -value, assuming an observed sample proportion of $\hat{p} = 0.603$ instead of 0.745. To show your work, include a screenshot below making sure your screenshot includes any/all inputs and output!

- My new approximate theory-based p -value is 0.0448

Solution:

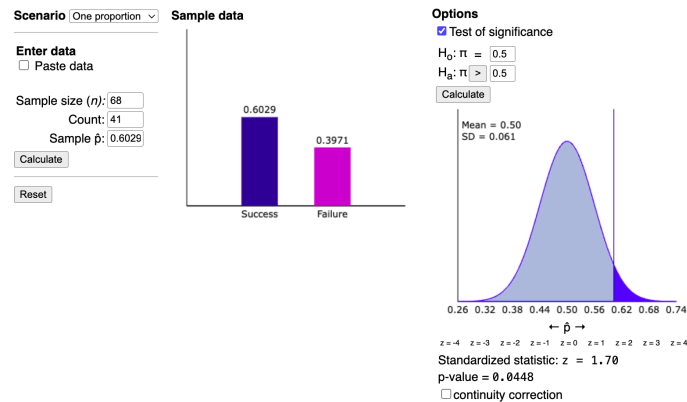


Figure 4: Results from the Theory-Based Inference, assuming the same null and alternative hypotheses but a $\hat{p}$ value of 0.603.



Question 3

- (a) Based on your result to Question 2 above, was your guess in Question 1(a) correct?

Solution: Answers may vary.

- (b) Based on your result to Question 2 above, was your guess in Question 1(b) correct?

Solution: Answers may vary.



Pro Tip

So, to summarize, the closer the observed statistic is to the average value of the null distribution the less evidence there is against the null hypothesis (and the larger the p -value).

(B) Sample Size

The second factor we will look at that influences strength of evidence against the null hypothesis is the sample size. The data for this study came from 68 statistics students. Suppose now we collected data from 25 students and 19 of them chose Tim on the left. The statistic (proportion of students choosing Tim on the left) of $19/25 = 0.760$ is very similar to to our original one (0.765). Let's see what the smaller sample size does to the strength of evidence.



Question 4

(a) How do you think changing the sample size to a smaller value (25 instead of 68) but leaving the value of $\hat{p}$ roughly unchanged will affect the approximate p -value?

- ☐ The approximate p -value will be smaller
- ☐ The approximate p -value will stay the same
- ☐ The approximate p -value will be larger

Solution: There isn't really a right or wrong answer to this part - it's just asking for your opinion!

(b) How do you think changing the sample size to a smaller value (25 instead of 68) but leaving the value of $\hat{p}$ roughly unchanged will affect the strength of evidence against the null hypothesis?

- ☐ The evidence will be stronger
- ☐ The evidence will stay the same
- ☐ The evidence will be weaker

Solution: Again, there isn't really a right or wrong answer to this part - it's just asking for your opinion! But, I do encourage you to make sure is consistent with your answer to the previous part - remember that the smaller the p -value, the stronger the evidence against the null (and, on the flipside, the larger the p -value, the weaker the evidence against the null)



Question 5

Navigate to the One Proportion applet. Using at least 10,000 repetitions, find the approximate standard deviation of the null distribution assuming we have the smaller sample size. To show your work, include a screenshot below making sure your screenshot includes any/all inputs and output!

Don't close out of your plot yet - we'll modify it in the next few questions.

- The approximate value for SD_{null} when $n = 68$ is **0.061**.
- The approximate value for SD_{null} when $n = 25$ is 0.101

Solution:

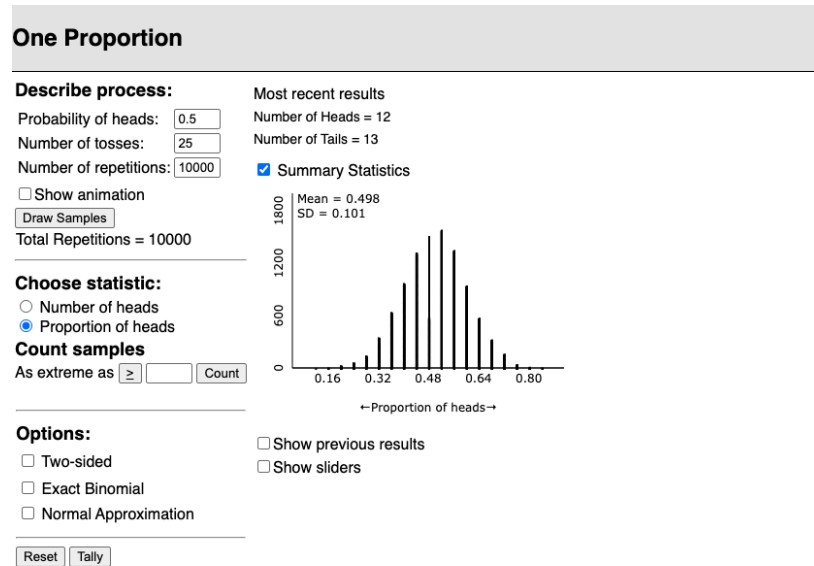


Figure 5: Results from the One-Proportion Applet, assuming a new smaller sample size of $n = 25$ (as opposed to the original $n = 68$).

You should have noticed that with a smaller sample size, SD_{null} increases. In other words, we see more variability in the null distribution:

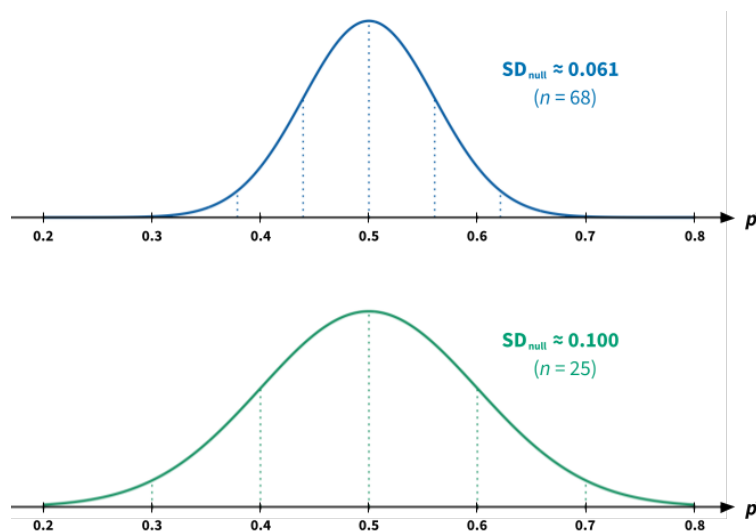


Figure 6: Illustration of how null distributions resulting from smaller sample sizes have larger standard deviations.

Even though the **central limit theorem** does not always apply (in particular, if the validity conditions are not satisfied), the standard deviation formula it provides can be insightful in the context of sample size:

$$SD_{null} \text{ from CLT} = \sqrt{\frac{\pi_0(1 - \pi_0)}{n}}$$

Because the sample size n appears in the denominator, as n gets larger the overall SD gets smaller - which is exactly what we saw empirically, above!

Question 6

Navigate back to your plot from the One Proportion Applet. Report both the simulation-based approximate p -value and the theory-based approximate p -value (**Hint:** “Summary Statistics”) under this new data ($n = 25$ students, 19 of whom assigned “Tim” to the left photo). To show your work, include a screenshot below making sure your screenshot includes any/all inputs and output!

- When $\hat{p}$ is roughly 0.765 but the sample size is $n = 68$, the approximate simulation-based p -value is **0.000**.
- When $\hat{p}$ is roughly 0.760 but the sample size is $n = 25$, the approximate **simulation**-based p -value is 0.0056
- When $\hat{p}$ is roughly 0.760 but the sample size is $n = 25$, the approximate **theory**-based p -value is 0.0047

Solution: Based on our answer to Question ??, we *must* use the simulation-based applet:

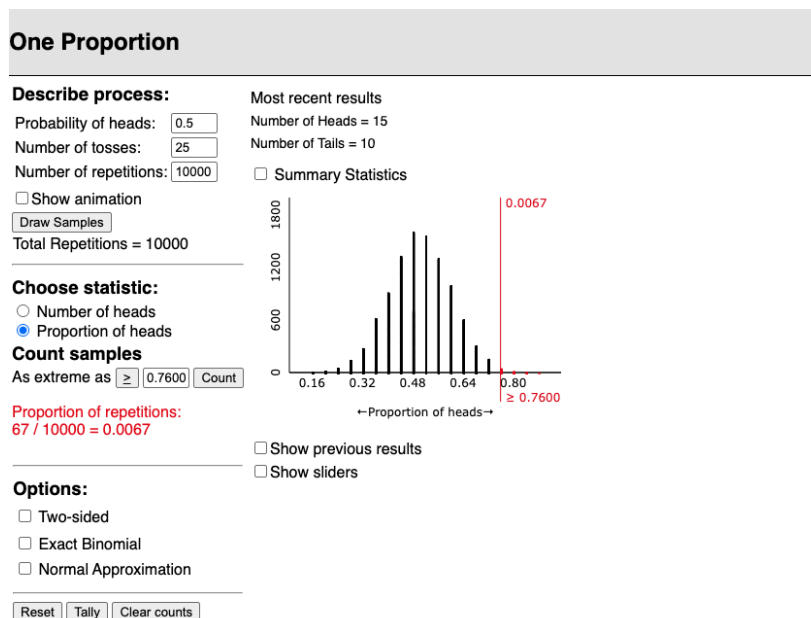


Figure 7: Results from the One-Proportion Applet, assuming a new smaller sample size of $n = 25$ (as opposed to the original $n = 68$).

Question 7

You should notice that your two p -values (from the simulation- and theory-based approaches) are a bit different this time. Which do you trust more, and why? (There **is** a correct answer here!)

Solution: This question is just a roundabout way of asking about the **validity conditions**. Note that in the new data setup we have only 6 failures, which is not greater than or equal to 10. Hence, **the validity conditions are NOT met** and as such **the central limit theorem is not guaranteed to apply**. As such, we should trust the **simulation-based approach** more.

 Question 8

Based on your answer to Question (6), were your answers to Question 4 correct?

Solution: Answers may vary.

 Pro Tip

As the sample size decreases, but the value of the observed sample statistic stays the same, the strength of evidence against the null hypothesis becomes weaker (larger p -value).

Part 2: One-Sided vs Two-Sided Tests

In this part of the activity, we are going to explore one-sided tests versus two-sided tests. Up until now we have only considered one-sided tests, where the alternative hypothesis uses either a $<$ or a $>$ sign. In a two-sided test, the alternative hypothesis uses the $\neq$ sign. So, why would we do a two-sided test and what are the implications?

Suppose the researchers did not necessarily think that people were more likely to put Tim *on the left*, but they also didn't necessarily think that Tim *on the right* was more likely. They were just interested in whether one "side" would be chosen more often than the other. A two-sided alternative hypothesis (the proportion of all my STAT 1110 students that put "Tim" on the left is **not equal to** 50%) allows the researchers to be less sure of the anticipated value of the parameter than a one-sided test.

Let's imagine we have a different dataset: out of 50 respondents from my STAT 1110 lecture sections, 30 assigned the name "Tim" to the photo on the left (so that $\hat{p} = 0.60$).

 Question 9

Report the null and alternative hypotheses for a two-sided test.

Solution: Let π_{TimLeft} denote the proportion of all of my STAT 1110 students that would assign the name "Tim" to the photo on the left. Then, a null hypothesis with a two-sided alternative that tests "there is no facial prototyping going on" against "there is facial prototyping going on" would be

$$\begin{cases} H_0 : \pi_{\text{TimLeft}} = 0.5 \\ H_A : \pi_{\text{TimLeft}} \neq 0.5 \end{cases}$$

 Question 10

Find and report the theory-based approximate p -value for the two-sided test. To show your work, include a screenshot of the entire screen showing any/all relevant inputs and outputs to and from

the applet.

- The approximate theory-based p -value for the two-sided test is 0.1573

Solution:

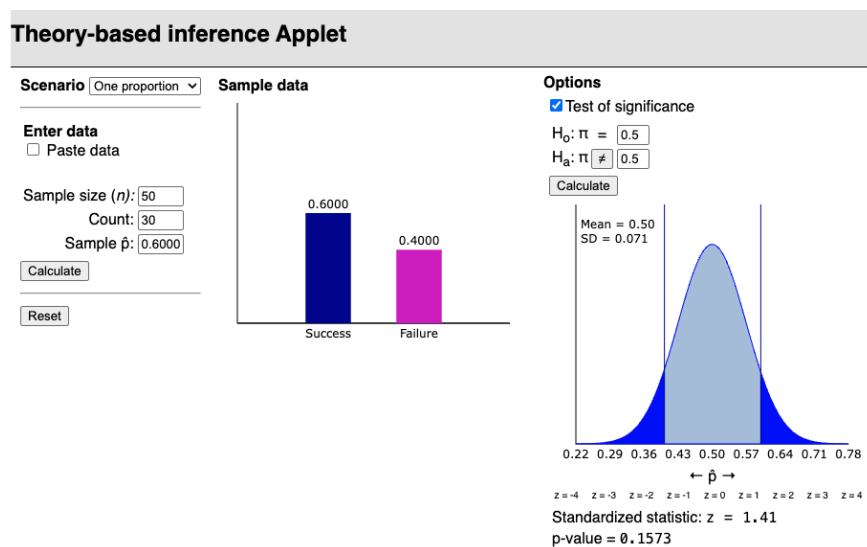


Figure 8: Results from the Theory-Based Inference, for $\hat{p} = 30/50$ and assuming a two-sided alternative.

Question 11

Keeping $\hat{p} = 30/50 = 0.6$ but using an upper-tailed alternative (i.e. an alternative of the form we used at the start of this activity) results in a theory-based p -value of 0.0786. Based on this, along with your answers to Question 10 above, what happens to the p -value if we switch from a one-sided test to a two-sided test?

Solution: The p -value increases, and the strength of evidence against the null becomes weaker.

Question 12

Interpret the two-sided p -value.

The p -value of 0.1573 is the probability of observing a sample proportion of students who put Tim's name on the left of **0.60** or **more extreme values on both tails** under the assumption that people are **equally likely** Tim on the left or right.

Hint: the second blank should be "greater values," "smaller values," or "more extreme values on both tails"

Caution

In a real-world problem, it is ill-advised to switch your alternative hypothesis partway through the analysis. It is always good practice to set your hypotheses before ever seeing any data, to ensure you are not unduly influenced by the data.

💡 Pro Tip

Because the p -value for a two-sided test is about twice as large as that for a one-sided test, they provide less evidence against the null hypothesis. However, two-sided tests are used more often in scientific practice.

Take it Further - Optional

If you feel adventurous or find these topics comfortable, explore the optional questions below. They are designed to stretch your thinking a bit further.

In the original study, the hypotheses were $H_0 : \pi_{\text{TimLeft}} = 0.50$ and $H_A : \pi_{\text{TimLeft}} > 0.50$, the statistic was 0.765, the z -statistic was 4.37, and the approximate (both simulation-based and theory-based) p -value was 0.000. Imagine a new study was conducted and the statistic for this new study was 0.51.

- 1) How would this have affected the p -value? The p -value for the new study would be ...

✓ **Larger** ☐ The same ☐ Smaller

- 2) Explain your answer to the previous question.

The statistic of 0.51 is closer to the null value of 0.50, so the p -value would increase as “more extreme values” in the calculation of the p -value would include all values greater than 0.51 (instead of greater than 0.765). This p -value is larger than the original study’s p -value of 0.000, so it gives less evidence against the null.

- 3) How would this have affected the absolute value of the standardized statistic? The absolute value of the standardized statistic for the new study would be ...

☐ Larger ☐ The same ✓ **Smaller**

- 4) Explain your answer to the previous question.

The standardized statistic measures how many standard deviations the statistic is from the null value. Since 0.51 is very close to 0.50 (much closer than 0.765), the absolute value of the z -statistic would be smaller.

- 5) How would this have affected the strength of evidence against the null hypothesis? The strength of evidence against the null hypothesis for the new study would be ...

☐ Stronger ☐ The same ✓ **Weaker**

- 6) Explain your answer to the previous question.

See answers to question (2) above.